

Multi-Agent Systems and Agentic AI

Practical Examination Report

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Team members for Part I: Name 1, Name 2

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Repository and logs. The code used for Part I is available at [GitLab repository](#). Training and evaluation logs are available at [W&B run](#). The adaptive-planning implementation studied in Part II is linked at [cmbagent_lg adaptive-planning branch](#). Replace these placeholders with exact clickable URLs, including the exact commit where required.

Notes on jointly owned work. Sections I.1–I.3 describe experiments run using the shared team TPU and shared repository. The analysis and prose in this report are my own. Section I.4 and Part II are completed individually.

Contents

Part I: GRPO finetuning of Gemma 3 on a TPU

I.1 Reproducing the baseline

Table 1: Baseline run details.

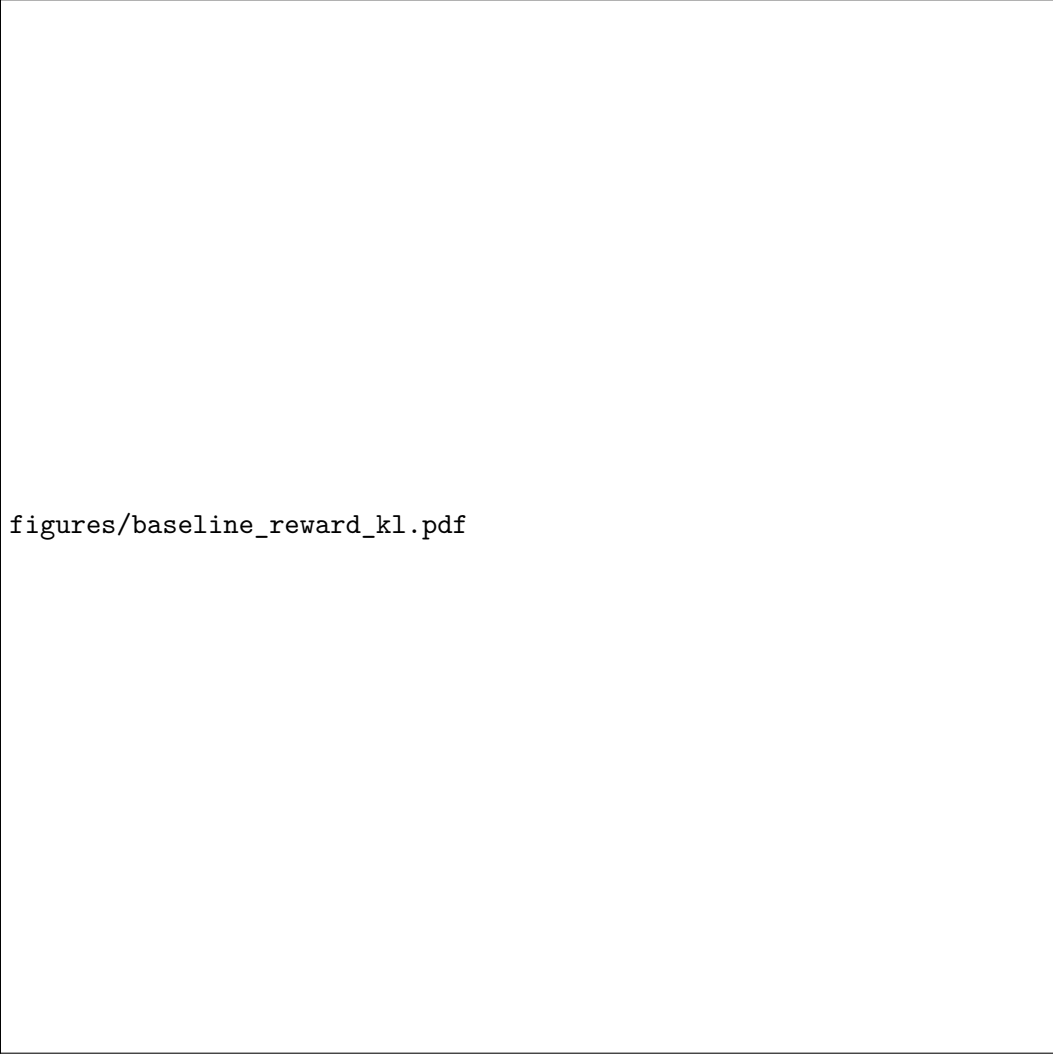
Item	Value
Repository commit	TODO
TPU type	v6e-1
Training command	<code>python scripts/train.py</code>
Evaluation command	<code>python scripts/evaluate.py</code>
Wall-clock time	[TODO: e.g. 6 h 20 min]
GRPO steps completed	[TODO: number of steps]
Held-out split	[TODO: e.g. GSM8K test split]
Evaluation seed	[TODO: fixed seed]

Run details.

Baseline patches. [TODO: State whether the baseline ran as shipped. If not, document the minimal fix required, with the affected file and commit. Keep this short.]

Table 2: Baseline GSM8K accuracy.

Model	Split	Seed	Accuracy
Base model, <code>google/gemma-3-1b-it</code>	[TODO: split]	[TODO: seed]	[TODO: value]
LoRA GRPO checkpoint	[TODO: split]	[TODO: seed]	[TODO: value]



figures/baseline_reward_kl.pdf

Figure 1: Baseline training curves showing mean reward $\bar{r}$ and $\text{KL}(\pi_\theta \parallel \pi_{\text{ref}})$ against GRPO step.

Evaluation.

I.2 Understanding the baseline

Policies in the LoRA setup. [TODO: Explain what plays the role of π_θ , π_{old} and π_{ref} , and identify which carries trainable parameters.]

Group size and advantage estimator. [TODO: State the group size K from `scripts/config.py`. Identify the Tunix advantage estimator and state whether it is standard group-mean or leave-one-out.]

Reward terms. [TODO: Identify shaping rewards and the true correctness reward in `scripts/rewards.py`. Explain what would happen early in training with correctness reward alone in terms of within-group reward variance σ_r .]

Clipped surrogate. [TODO: Locate where the PPO-style clipped surrogate is implemented in Tunix. State what ε corresponds to in `scripts/config.py` and note any deviation from the lecture form, such as per-token rather than per-sequence ratios.]

I.3 Improving on the baseline

Modification and motivation. [TODO: Describe the principled modification. Examples: leave-one-out advantage, sweep K , change β or ε , modify shaping rewards, add length penalty, change LoRA

rank, learning rate, schedule or generation length. Connect the choice to GRPO theory.]

Controlled comparison. [TODO: State how the comparison was controlled: same data, same evaluation split, same total compute or normalised per step. Mention baseline plus at least two further complete runs, or justify fewer.]

Table 3: Controlled comparison on held-out GSM8K. Include uncertainty using a second seed or a bootstrap confidence interval over evaluation prompts.

Model / run	Modification	Seed(s)	Accuracy	Uncertainty
Base model	None	[TODO:]	[TODO:]	[TODO:]
Baseline GRPO	Default config	[TODO:]	[TODO:]	[TODO:]
Variant 1	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Variant 2	[TODO:]	[TODO:]	[TODO:]	[TODO:]

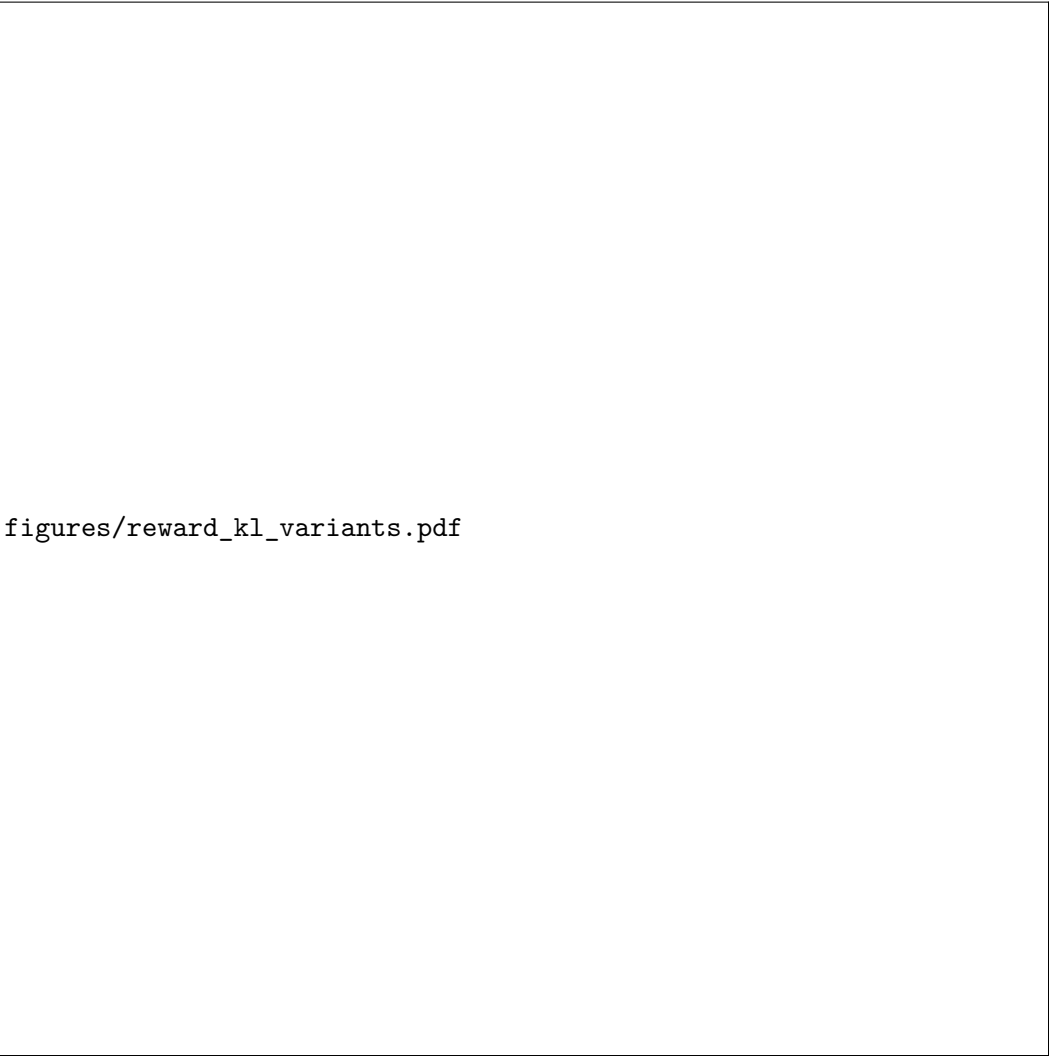


Figure 2: Training curves for the baseline and variants on shared axes, showing mean reward and KL over GRPO step.

figures/grpo_diagnostic.pdf

Figure 3: Diagnostic plot for a known GRPO failure mode, such as advantage distribution, degenerate reward groups, response length, entropy, or KL versus accuracy.

Discussion. [TODO: Connect the empirical findings to lecture theory. State whether the observed behaviour matched the prediction, and diagnose any negative results honestly.]

I.4 GRPO theory

Question 1

(i) Since $\mathbb{E}[\bar{r}] = \mu$,

$$\mathbb{E}[\hat{A}_i] = \frac{\mathbb{E}[r_i] - \mathbb{E}[\bar{r}]}{\sigma} = 0, \quad \mathbb{E}[X_i] = h_i \mathbb{E}[\hat{A}_i] = 0, \quad \mathbb{E}[\hat{g}] = 0.$$

Here the responses, and hence their score vectors, are fixed; the expectation is only over independent reward noise. In the policy-gradient expectation the responses are instead sampled from the policy, so response quality and score can be correlated and the gradient need not vanish.

(ii) Writing

$$X_i = \frac{h_i}{\sigma} r_i - \frac{h_i}{K\sigma} \sum_{\ell=1}^K r_\ell$$

exhibits the shared random term, namely the group-mean contribution $-h_i\bar{r}/\sigma$. For $i \neq j$, independence of the rewards gives

$$\begin{aligned}\text{Cov}(r_i - \bar{r}, r_j - \bar{r}) &= 0 - \frac{\sigma^2}{K} - \frac{\sigma^2}{K} + \frac{\sigma^2}{K} = -\frac{\sigma^2}{K}, \\ \text{Cov}(X_i, X_j) &= -\frac{1}{K} h_i h_j^\top.\end{aligned}$$

Thus the cross-covariance has a negative coefficient: an unusually large r_i raises $\bar{r}$, thereby reducing every other normalised advantage.

(iii) The diagonal terms are $\text{Var}(X_i) = (1 - K^{-1})h_i h_i^\top$. Therefore, with $\bar{h} = K^{-1} \sum_i h_i$,

$$\text{Var}(\hat{g}) = \frac{1}{K^2} \left[\sum_i h_i h_i^\top - \frac{1}{K} \left(\sum_i h_i \right) \left(\sum_j h_j \right)^\top \right] = \frac{1}{K^2} \sum_i (h_i - \bar{h})(h_i - \bar{h})^\top. \quad (1)$$

This is generally not the naive i.i.d. value $K^{-1} \text{Var}(X_1)$. Because these are covariance matrices, an effective sample size requires a scalar projection. In a direction u define

$$K_{\text{eff}}(u) := \frac{u^\top \text{Var}(X_1) u}{u^\top \text{Var}(\hat{g}) u} = \frac{K(K-1)(u^\top h_1)^2}{\sum_i \{u^\top (h_i - \bar{h})\}^2}, \quad (2)$$

with $K_{\text{eff}}(u) = \infty$ when the denominator vanishes. A rotation-invariant trace summary is

$$K_{\text{eff, tr}} := \frac{\text{tr Var}(X_1)}{\text{tr Var}(\hat{g})} = \frac{K(K-1)\|h_1\|^2}{\sum_i \|h_i - \bar{h}\|^2}.$$

For comparable, non-degenerate score vectors, the denominator grows as $O(K)$ and $K_{\text{eff}} = O(K)$, although its constant need not equal the i.i.d. one. If $h_i = c_i v$ are all aligned, $??$ is rank one and only variation in the c_i remains; if all h_i are equal, $\hat{g} = h_1 K^{-1} \sum_i \hat{A}_i = 0$ exactly.

Question 2

(i) Assume $\pi_t(a) > 0$ on the admissible support and, in the measurable case, that the normaliser below is finite. Let λ enforce $\sum_a \pi(a) = 1$. Stationarity of the Lagrangian gives

$$A_t(a) - \frac{1}{\eta} \left(\log \frac{\pi(a)}{\pi_t(a)} + 1 \right) + \lambda = 0.$$

Normalising the resulting exponential tilt yields the unique optimiser

$$\pi_{t+1}(a) = \frac{\pi_t(a) \exp\{\eta A_t(a)\}}{Z_t}, \quad Z_t = \sum_b \pi_t(b) \exp\{\eta A_t(b)\}. \quad (3)$$

Strict concavity from the negative KL term gives uniqueness. Small η keeps π_{t+1} close to π_t ; increasing η weakens the KL regularisation and makes a larger step towards high-advantage actions.

(ii) Let $F_t(\pi) = \mathbb{E}_{a \sim \pi}[A^{\pi_t}(a)] - \eta^{-1} \text{KL}(\pi \| \pi_t)$. The defining property of the true advantage is $\mathbb{E}_{a \sim \pi_t}[A^{\pi_t}(a)] = 0$, so $F_t(\pi_t) = 0$. Optimality of $??$ then implies

$$J(\pi_{t+1}) = \mathbb{E}_{\pi_{t+1}}[A^{\pi_t}] \geq \frac{1}{\eta} \text{KL}(\pi_{t+1} \| \pi_t) \geq 0 = J(\pi_t).$$

Thus the exact unrestricted update is monotone. In neural-network GRPO the group-normalised advantage is a noisy approximation and stochastic optimisation need not find the exact distributional maximiser, so the guarantee can fail.

(iii) For $\hat{A}_t > 0$, clipping removes the incentive to increase the sampled ratio beyond $1 + \varepsilon$; for $\hat{A}_t < 0$, it removes the incentive to decrease it below $1 - \varepsilon$. This is an advantage-dependent, samplewise *soft* trust region: it does not enforce either bound, since shared parameter updates or other samples can still move a ratio outside the interval. By contrast, $\text{KL}(\pi_\theta(\cdot | q) \| \pi_{\text{old}}(\cdot | q)) \leq \delta$ is one hard, distribution-level constraint; it can permit a large change on some actions provided the joint KL budget remains satisfied.

Question 3

(i) For any integrand f , $\mathbb{E}_{\pi_{\text{old}}}[f] = \mathbb{E}_{\pi_{\text{mix}}}[wf]$, where

$$w_i = \frac{\pi_{\text{old}}(a_i | q)}{\pi_{\text{mix}}(a_i | q)} = \frac{\pi_{\text{old}}(a_i | q)}{(1 - \alpha)\pi_{\text{old}}(a_i | q) + \alpha\pi_{\text{unif}}(a_i | q)}. \quad (4)$$

Thus the direct importance-corrected version of the stated GRPO estimator is

$$\hat{g}_{\text{mix}} = \frac{1}{K} \sum_{i=1}^K w_i \nabla_{\theta} \log \pi_{\theta}(a_i | q) \frac{r_i - \bar{r}}{\sigma_r}, \quad a_i \stackrel{\text{iid}}{\sim} \pi_{\text{mix}}. \quad (5)$$

(ii) For each fixed realised action, $\pi_{\text{mix}}(a_i | q) \rightarrow \pi_{\text{old}}(a_i | q)$ and hence $w_i \rightarrow 1$. The realised r_i , $\bar{r}$, and σ_r are unchanged, so ?? tends to $\hat{g}_{\text{GRPO}}$.

(iii) Linearity of expectation gives

$$V^{\pi_{\text{mix}}}(q) = (1 - \alpha)V^{\pi_{\text{old}}}(q) + \alpha V^{\pi_{\text{unif}}}(q).$$

Consequently, multiplying by w_i outside the advantage does not change the fact that the unweighted $\bar{r}$ estimates $V^{\pi_{\text{mix}}}$ rather than $V^{\pi_{\text{old}}}$. At the special point $\pi_{\theta} = \pi_{\text{old}}$, an independent constant baseline cancels because $\mathbb{E}_{\pi_{\text{old}}}[\nabla \log \pi_{\text{old}}] = 0$; the random same-group $\bar{r}$ is not independent, however. A common consistent correction is

$$\hat{V}_{\text{SN}} = \frac{\sum_j w_j r_j}{\sum_j w_j}, \quad \tilde{A}_i = \frac{r_i - \hat{V}_{\text{SN}}}{\sigma_r},$$

which also reduces to the stated GRPO advantage when $\alpha \rightarrow 0$. However, self-normalisation and reusing sample i in its own baseline cause finite-sample bias. For an exactly unbiased baseline correction, use a fixed or independently estimated scale s and the leave-one-out importance baseline

$$\hat{V}_{-i} = \frac{1}{K-1} \sum_{j \neq i} w_j r_j, \quad \tilde{A}_i^{\text{LOO}} = \frac{r_i - \hat{V}_{-i}}{s}. \quad (6)$$

Indeed, $\mathbb{E}_{\pi_{\text{mix}}}[\hat{V}_{-i}] = V^{\pi_{\text{old}}}$ and $\hat{V}_{-i}$ is independent of a_i . Therefore substituting ?? into ?? is unbiased for the corresponding fixed-scale π_{old} score-advantage expectation. Using the same group's empirical σ_r retains the usual finite-sample normalisation bias.

(iv) First, the unweighted reward variance under the proposal is

$$\begin{aligned} \text{Var}_{\pi_{\text{mix}}}(R) &= (1 - \alpha) \text{Var}_{\pi_{\text{old}}}(R) + \alpha \text{Var}_{\pi_{\text{unif}}}(R) \\ &\quad + \alpha(1 - \alpha)(V^{\pi_{\text{old}}} - V^{\pi_{\text{unif}}})^2. \end{aligned}$$

It therefore increases when uniform responses have larger reward variance, or a sufficiently different mean; for example, when uniform sampling exposes many extreme invalid and accidentally correct responses. It decreases when $V^{\pi_{\text{unif}}} \simeq V^{\pi_{\text{old}}}$ and $\text{Var}_{\pi_{\text{unif}}}(R) < \text{Var}_{\pi_{\text{old}}}(R)$.

Importance weighting adds a second effect. For a scalar projection Y of the score-advantage term,

the leading per-sample second moment is

$$\mathbb{E}_{\pi_{\text{mix}}}[w^2 Y^2] = \mathbb{E}_{\pi_{\text{old}}}[w Y^2].$$

Thus mixing increases estimator variance if large $|Y|$ occurs where $\pi_{\text{old}} > \pi_{\text{mix}}$ (so $w > 1$), but can decrease it if the uniform component moves proposal mass towards large- $|Y|$ responses that π_{old} under-samples (so $w < 1$ there). Hence lower raw advantage variance alone does not guarantee lower variance of $\hat{g}_{\text{mix}}$.

(v) Under the stated assumption,

$$w_i = c^T = \exp(T \log c) \longrightarrow 0 \quad \text{exponentially as } T \rightarrow \infty.$$

When $\alpha > 0$, mixture samples include tokens to which the uniform component assigns more probability than π_{old} ; more formally, $\mathbb{E}_{\pi_{\text{mix}}}[\log(\pi_{\text{old}}/\pi_{\text{mix}})] = -\text{KL}(\pi_{\text{mix}}\|\pi_{\text{old}}) < 0$, so the geometrically typical per-token ratio is below one. Consequently most long responses contribute almost nothing while a few rare, large-weight responses dominate, giving a very small effective sample size and an unreliable gradient. One concrete fix is to scale the mixture coefficient as $\alpha = \gamma/T$, keeping the cumulative proposal perturbation, and hence typical log-weight magnitude, $O(1)$; the trade-off is progressively weaker uniform exploration for longer responses.

Part II: Adaptive planning in `cmbagent_lg`

II.1 Workflow diagram

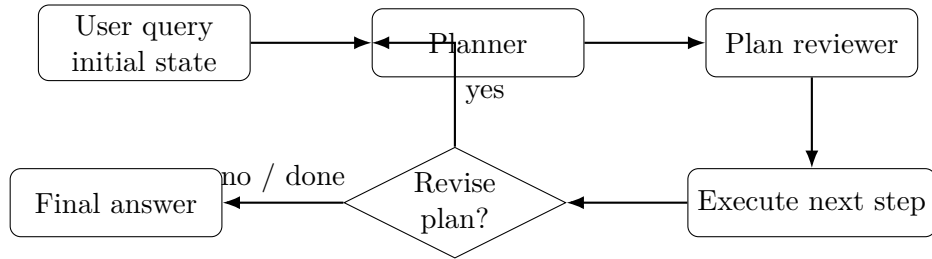


Figure 4: Placeholder adaptive-planning workflow. Replace this with the exact LangGraph nodes, edges, conditional routing, and state fields from the studied commit. This diagram does not count towards the 2-page Part II limit.

Walkthrough. [TODO: In at most half a page, explain what triggers a plan revision, what part of the plan may change, what is held fixed across a revision, and how the loop terminates. Include the exact commit link studied.]

II.2 Problems

Problem 1: [TODO: short name]. [TODO: Argue the limitation or failure mode, grounded in planning and closed-loop control theory.]

Problem 2: [TODO: short name]. [TODO: Argue the limitation or failure mode, grounded in planning and closed-loop control theory.]

Problem 3: [TODO: short name]. [TODO: Argue the limitation or failure mode, grounded in planning and closed-loop control theory.]

II.3 Proposed improvements

Improvement 1. [TODO: State what changes, why it addresses the corresponding problem, and what it trades off.]

Improvement 2. [TODO: State what changes, why it addresses the corresponding problem, and what it trades off.]

Improvement 3. [TODO: State what changes, why it addresses the corresponding problem, and what it trades off.]

References

- Hu, E. J. et al. (2021). *LoRA: Low-Rank Adaptation of Large Language Models*. <https://arxiv.org/abs/2106.09685>
- Tunix documentation or source link used for GRPO implementation: <https://github.com/google/tunix>
- Exact cmbagent_lg adaptive-planning commit studied: [TODO: insert clickable commit URL]

A Appendix

A.1 Additional run details

[TODO: Optional non-essential logs, extra tables, hyperparameter dumps, and supplementary figures.]